Gurgaon Residential Real Estate Market Analysis: Understanding Pricing Drivers and Identifying Value Opportunities

Executive Overview

Gurgaon is one of India's most active and rapidly evolving residential real estate markets, characterized by significant variation in property prices, housing configurations, locality performance, and investment potential. For buyers and investors, evaluating a property based solely on its listed price often provides an incomplete picture of its true value.

This project analyzes **10,000+ residential property listings across 220+ Gurgaon localities** to understand the factors that influence property valuations, identify premium and value-oriented micro-markets, and evaluate how property characteristics and location interact to shape housing prices.

The original dataset contained both **property sale listings and rental listings**. To ensure analytical consistency and meaningful price comparisons, the dataset was segmented into separate sales and rental datasets. All subsequent analysis in this project focuses exclusively on the **residential sales market**, allowing for a more accurate evaluation of pricing dynamics, locality performance, and investment opportunities.

Through a combination of data cleaning, missing-value treatment, feature engineering, anomaly validation, and exploratory analysis, the project transforms raw housing listings into actionable market intelligence that can support smarter home-buying and investment decisions.

Business Problem

Residential property decisions are often made using limited information, with buyers focusing primarily on total property price while overlooking important factors such as locality performance, price-per-square-foot efficiency, property size, configuration, and market positioning.

As a result, buyers frequently struggle to answer critical questions such as:

- Which factors have the greatest influence on property prices?
- Which localities command genuine pricing power?
- Where can buyers find better space-to-price value?
- How do luxury markets differ from the broader housing market?
- Which localities offer the strongest balance between affordability, space, and long-term potential?

This project addresses these questions through a structured analysis of Gurgaon's residential property sales market.

Analytical Objectives

The analysis focuses on five key areas:

1. Market Structure Analysis

Understand the composition of Gurgaon's residential housing market through inventory distribution, pricing patterns, and property configurations.

2. Pricing Driver Analysis

Identify the property attributes that most strongly influence housing prices and evaluate how area, configuration, and other characteristics contribute to valuation.

3. Locality Performance Analysis

Compare Gurgaon localities based on inventory concentration, pricing levels, and price-per-square-foot performance.

4. Luxury Market Profiling

Investigate the characteristics that distinguish premium residential properties from the broader housing market.

5. Value Opportunity Identification

Identify localities that provide favorable combinations of living space, pricing efficiency, and investment potential.

Dataset Overview

Source: 99acres Residential Property Listings

Scope of Analysis

- 10,000+ Residential Property Listings
- 220+ Gurgaon Localities
- Sale and Rental Listings Available in the Original Dataset
- Apartments, Builder Floors, and Independent Houses
- Property Attributes Including Area, Price, BHK Configuration, Bathrooms, Amenities, Furnishing, Property Type, and Locality Information

Data Preparation

The original dataset was first segmented into separate **sales** and **rental** datasets. Since sale and rental properties follow fundamentally different pricing dynamics, combining them would distort valuation metrics, pricing distributions, and market insights. Therefore, all exploratory analysis was conducted exclusively on the sales dataset.

The data then underwent a structured preparation process that included:

- Missing-value treatment
- Feature engineering
- Data-quality validation
- Area-based anomaly detection
- Price-per-square-foot anomaly treatment

Properties with unrealistic area values and extreme lower-tail price-per-square-foot observations were identified and removed to improve analytical reliability and ensure that the final dataset accurately reflected Gurgaon's residential sales market.

The resulting analytical dataset provides a robust foundation for understanding pricing behavior, locality performance, luxury market dynamics, and value-oriented investment opportunities within Gurgaon's residential housing market.

Import Libraries

In [7]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

pd.set_option('display.max_columns', None)
```

Data Loading

In [8]:
```
data_frame= pd.read_csv(r"D:\DATA analysis\projects\Key Projects\99 acre\Gurgaon
df=data_frame.copy()
df.head()
```

Out[8]:

	PROP_ID	PHOTO_URL	
0	O71585872	https://mediacdn.99acres.com/media1/22409/13/4...	https://mediacdn.99acres.cc
1	T71585466	https://mediacdn.99acres.com/media1/22425/6/44...	https://mediacdn.99acres.cc
2	D71549524	https://mediacdn.99acres.com/media1/22386/13/4...	https://mediacdn.99acres.cc
3	I71493958	https://mediacdn.99acres.com/media1/22353/14/4...	https://mediacdn.99acres.cc
4	H71479412	https://mediacdn.99acres.com/media1/22352/4/44...	https://mediacdn.99acres.cc

Initial Data Exploration

In [9]:

```
df.shape
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10704 entries, 0 to 10703
Data columns (total 67 columns):
 #   Column                         Non-Null Count  Dtype
---  ------                         --------------  -----
 0   PROP_ID                        10704 non-null  object
 1   PHOTO_URL                      8864 non-null   object
 2   MEDIUM_PHOTO_URL               8864 non-null   object
 3   PREFERENCE                     10704 non-null  object
 4   DESCRIPTION                    10704 non-null  object
 5   PROPERTY_TYPE                  10704 non-null  object
 6   CITY                           10704 non-null  object
 7   LOCALITY                       10704 non-null  object
 8   TRANSACT_TYPE                  10152 non-null  float64
 9   OWNTYPE                        10704 non-null  int64
 10  BEDROOM_NUM                    10511 non-null  float64
 11  BATHROOM_NUM                   10704 non-null  int64
 12  BALCONY_NUM                    10501 non-null  float64
 13  PRICE_PER_UNIT_AREA            10704 non-null  float64
 14  FURNISH                        10704 non-null  int64
 15  FACING                         10704 non-null  int64
 16  AGE                            10704 non-null  int64
 17  FLOOR_NUM                      9804 non-null   object
 18  TOTAL_FLOOR                    10697 non-null  float64
 19  FEATURES                       10704 non-null  object
 20  REGISTER_DATE                  10704 non-null  object
 21  PROP_NAME                      9135 non-null   object
 22  MIN_PRICE                      10704 non-null  int64
 23  MAX_PRICE                      10704 non-null  int64
 24  PRICE_SQFT                     10704 non-null  float64
 25  LISTING                        10704 non-null  object
 26  BUILDING_ID                    10704 non-null  int64
 27  CARPET_SQFT                    6539 non-null   float64
 28  SUPERBUILTUP_SQFT              6712 non-null   float64
 29  BROKERAGE                      10623 non-null  float64
 30  MAP_DETAILS                    10704 non-null  object
 31  FSL_Data                       10704 non-null  object
 32  MIN_AREA_SQFT                  10704 non-null  float64
 33  MAX_AREA_SQFT                  10704 non-null  float64
 34  FORMATTED                      10704 non-null  object
 35  AMENITIES                      10620 non-null  object
 36  TOP_USPS                       8179 non-null   object
 37  PD_URL                         10702 non-null  object
 38  EXPIRY_DATE                    10704 non-null  object
 39  GROUP_NAME                     10704 non-null  object
 40  AREA                           10704 non-null  object
 41  PRICE                          10704 non-null  object
 42  PROP_HEADING                   10704 non-null  object
 43  PROP_DETAILS_URL               10704 non-null  object
 44  CLASS_HEADING                  10704 non-null  object
 45  CLASS_LABEL                    10704 non-null  object
 46  SECONDARY_TAGS                 10704 non-null  object
 47  PROPERTY_IMAGES                8863 non-null   object
 48  THUMBNAIL_IMAGES               8863 non-null   object
 49  TOTAL_LANDMARK_COUNT           10638 non-null  float64
 50  FORMATTED_LANDMARK_DETAILS     10638 non-null  object
 51  CONTACT_NAME                   10660 non-null  object
 52  CONTACT_COMPANY_NAME           10699 non-null  object
 53  DEALER_PHOTO_URL               10704 non-null  object
 54  SOCIETY_NAME                   9135 non-null   object
```

```
 55  BUILDING_NAME                 9135 non-null   object
 56  CITY_ID                       10704 non-null  int64
 57  LOCALITY_WO_CITY              10704 non-null  object
 58  profile                       10704 non-null  object
 59  xid                           10704 non-null  object
 60  metadata                      10704 non-null  object
 61  location                      10704 non-null  object
 62  BUILTUP_SQFT                  2733 non-null   float64
 63  SUPER_SQFT                    875 non-null    float64
 64  COMMON_FURNISHING_ATTRIBUTES  0 non-null      float64
 65  QUALITY_SCORE                 2 non-null      float64
 66  FURNISHING_ATTRIBUTES         2 non-null      object
dtypes: float64(16), int64(9), object(42)
memory usage: 5.5+ MB
```

Out[9]:

	TRANSACT_TYPE	OWNTYPE	BEDROOM_NUM	BATHROOM_NUM	BALCONY_N
count	10152.000000	10704.000000	10511.000000	10704.000000	10501.000
mean	1.001182	0.968610	3.327181	3.484118	2.930
std	0.034362	0.304357	1.288234	1.518250	0.916
min	1.000000	0.000000	1.000000	0.000000	0.000
25%	1.000000	1.000000	3.000000	3.000000	2.000
50%	1.000000	1.000000	3.000000	3.000000	3.000
75%	1.000000	1.000000	4.000000	4.000000	4.000
max	2.000000	4.000000	22.000000	22.000000	4.000

Feature Selection

The original dataset contains 67 columns covering various aspects of property listings. However, not all variables are required for the objectives of this analysis.

To improve analytical efficiency, reduce noise, and focus on the most business-relevant information, a subset of key features is selected. The retained columns capture the essential dimensions of the property market, including:

- Property identification
- Location information
- Property characteristics
- Size and pricing metrics
- Furnishing and property condition
- Property descriptions and amenities
- Society/building information

By narrowing the dataset to these core variables, we create a cleaner and more focused analytical dataset that is easier to explore, visualize, and derive meaningful insights from.

In [11]:
```
selected_cols = [
    # Core identifiers
```

```
    'PROP_ID','PROP_NAME','PREFERENCE' ,

    # Location
    'CITY', 'LOCALITY',

    # Property details
    'PROPERTY_TYPE', 'BEDROOM_NUM', 'BATHROOM_NUM',
    'BALCONY_NUM', 'FLOOR_NUM', 'TOTAL_FLOOR',

    # Size & pricing
    'AREA', 'PRICE',

    # Additional features
    'FURNISH', 'FACING', 'AGE','BROKERAGE', 'CONTACT_COMPANY_NAME','CLASS_LABEL'

    # Society / building
    'SOCIETY_NAME'
]

df = df[selected_cols]
```

In [12]:
```
print(df.head())
```

```
     PROP_ID                 PROP_NAME PREFERENCE     CITY  \
0  O71585872  Alpha Corp GurgaonOne 84          S  Gurgaon
1  T71585466            DLF The Ultima          S  Gurgaon
2  D71549524  Emaar MGF The Palm Drive          R  Gurgaon
3  I71493958       Experion Windchants          S  Gurgaon
4  H71479412               ATS Triumph          R  Gurgaon

             LOCALITY          PROPERTY_TYPE  BEDROOM_NUM  BATHROOM_NUM  \
0   Sector 84 Gurgaon  Residential Apartment          4.0             4
1   Sector 81 Gurgaon  Residential Apartment          4.0             4
2   Sector 66 Gurgaon  Residential Apartment          3.0             3
3  Sector 112 Gurgaon  Residential Apartment          3.0             3
4  Sector 104 Gurgaon  Residential Apartment          3.0             4

   BALCONY_NUM FLOOR_NUM  TOTAL_FLOOR         AREA    PRICE  FURNISH  FACING  \
0          4.0        14         26.0  3434 sq.ft.  2.63 Cr        4       1
1          3.0         7         29.0  2870 sq.ft.   3.6 Cr        4       1
2          2.0         7         17.0  1950 sq.ft.   85,000        1       4
3          3.0        14         28.0  2802 sq.ft.  3.85 Cr        4       0
4          4.0        12         27.0  2290 sq.ft.   40,000        4       6

   AGE  BROKERAGE CONTACT_COMPANY_NAME CLASS_LABEL             SOCIETY_NAME
0    2   263000.0         propertyline      Dealer  Alpha Corp GurgaonOne 84
1    1        1.0   MVD AND ASSOCIATES      Dealer            DLF The Ultima
2    1    42000.0    1st Nest Realtors      Dealer  Emaar MGF The Palm Drive
3    1        1.0            GROW MORE      Dealer       Experion Windchants
4    1        1.0            GROW MORE      Dealer               ATS Triumph
```

Dropped Columns:

- Image URLs → not useful for analysis
- Duplicate area columns → redundancy
- Metadata / JSON fields → complex, not needed
- Contact info → irrelevant

Fixing Datatypes

In [13]:
```
df.dtypes
```

Out[13]:
```
PROP_ID                 object
PROP_NAME               object
PREFERENCE              object
CITY                    object
LOCALITY                object
PROPERTY_TYPE           object
BEDROOM_NUM            float64
BATHROOM_NUM             int64
BALCONY_NUM            float64
FLOOR_NUM               object
TOTAL_FLOOR            float64
AREA                    object
PRICE                   object
FURNISH                  int64
FACING                   int64
AGE                      int64
BROKERAGE              float64
CONTACT_COMPANY_NAME    object
CLASS_LABEL             object
SOCIETY_NAME            object
dtype: object
```

In [14]:
```
df['FLOOR_NUM'] = pd.to_numeric(df['FLOOR_NUM'], errors='coerce')
```

Standardizing Area Column

In [15]:
```
df['AREA'] = df['AREA'].str.replace('sq.ft.', '', regex=False)
df['AREA'] = df['AREA'].str.replace(',', '')
df['AREA'] = pd.to_numeric(df['AREA'], errors='coerce')
```

Standardizing Price Column

In [16]:
```
def clean_price(x):
    x = str(x).lower().replace(',', '').replace('₹', '').strip()

    if x in ['price on request', 'call for price', '', 'nan', '—']:
        return None

    try:
        if 'cr' in x:
            return float(x.replace('cr', '').strip()) * 10000000
        elif 'lac' in x:
            return float(x.replace('lac', '').strip()) * 100000
        else:
            return float(x)
    except:
        return None
```

In [17]:
```
df['PRICE'] = df['PRICE'].apply(clean_price)
```

Data Cleaning

The PRICE and AREA columns contained mixed units and text values (e.g., "2.63 Cr", "3434 sq.ft.").

To ensure accurate analysis:

- Extracted numeric values
- Standardized price into INR
- Removed unit text from area

This step was critical to avoid incorrect aggregations.

Initial Missing Value Audit

In [19]:
```
df.isnull().sum().sort_values(ascending=False)
```

Out[19]:
```
PROP_NAME              1569
SOCIETY_NAME           1569
FLOOR_NUM              1326
PRICE                  1186
BALCONY_NUM             203
BEDROOM_NUM             193
BROKERAGE                81
TOTAL_FLOOR               7
CONTACT_COMPANY_NAME      5
AREA                      5
PROP_ID                   0
PREFERENCE                0
CITY                      0
PROPERTY_TYPE             0
BATHROOM_NUM              0
LOCALITY                  0
AGE                       0
FACING                    0
FURNISH                   0
CLASS_LABEL               0
dtype: int64
```

Handling Null Values For the Text Columns

PROP_NAME (Nulls - 1569)

Replaced nulls by 'unknown'

In [20]:
```
df['PROP_NAME'] = df['PROP_NAME'].fillna('Unknown')
```

SOCIETY_NAME (Nulls - 1569)

Replaced nulls by 'unknown'

In [21]:
```
df['SOCIETY_NAME'] = df['SOCIETY_NAME'].fillna('Unknown')
```

CONTACT_COMPANY_NAME (Nulls - 5)

Replaced nulls by 'unknown'

In [22]:
```
df['CONTACT_COMPANY_NAME'] = df['CONTACT_COMPANY_NAME'].fillna('Unknown')
```

In [23]:
```
df.isnull().sum().sort_values(ascending=False)
```

Out[23]:
```
FLOOR_NUM              1326
PRICE                  1186
BALCONY_NUM             203
BEDROOM_NUM             193
BROKERAGE                81
TOTAL_FLOOR               7
AREA                      5
PROP_ID                   0
CITY                      0
PREFERENCE                0
PROP_NAME                 0
BATHROOM_NUM              0
PROPERTY_TYPE             0
LOCALITY                  0
FACING                    0
FURNISH                   0
AGE                       0
CONTACT_COMPANY_NAME      0
CLASS_LABEL               0
SOCIETY_NAME              0
dtype: int64
```

Create Sales Dataset

In [25]:
```
df_sales = df[df['PREFERENCE'] == 'S'].copy()
```

Create Rental Dataset

In [26]:
```
df_rent = df[df['PREFERENCE'] == 'R'].copy()
```

In [28]:
```
print("Original Dataset :", len(df))
print("Sales Listings   :", len(df_sales))
print("Rental Listings  :", len(df_rent))
print("Check Total      :", len(df_sales) + len(df_rent))
```

```
Original Dataset : 10704
Sales Listings   : 10155
Rental Listings  : 549
Check Total      : 10704
```

Market Segmentation: Sales vs Rental Listings

The dataset contains both property sale and rental listings. Since pricing behavior, valuation metrics, and buyer objectives differ significantly between these two markets, combining them can lead to misleading insights.

To ensure analytical consistency, the dataset was segmented into separate sales and rental datasets.

SALES DATASET - MISSING VALUE TREATMENT

In [29]:
```
df_sales.isnull().sum()[df_sales.isnull().sum() > 0]
```

Out[29]:
```
BEDROOM_NUM     191
BALCONY_NUM     201
FLOOR_NUM      1269
TOTAL_FLOOR       7
AREA              5
PRICE          1077
BROKERAGE        72
dtype: int64
```

In [31]:
```
# PRICE
df_sales['PRICE'] = (
    df_sales['PRICE']
    .fillna(df_sales['PRICE'].median())
)


# AREA
df_sales = df_sales.dropna(subset=['AREA'])

# Bedroom Count
df_sales['BEDROOM_NUM'] = (
    df_sales['BEDROOM_NUM']
    .fillna(df_sales['BEDROOM_NUM'].mode()[0])
)

# Balcony Count
df_sales['BALCONY_NUM'] = (
    df_sales['BALCONY_NUM']
    .fillna(0)
)

# Floor Number
df_sales['FLOOR_NUM'] = (
    df_sales['FLOOR_NUM']
    .fillna(df_sales['FLOOR_NUM'].median())
)

# Total Floors
df_sales['TOTAL_FLOOR'] = (
    df_sales['TOTAL_FLOOR']
    .fillna(df_sales['TOTAL_FLOOR'].median())
)

# Brokerage
df_sales['BROKERAGE'] = (
    df_sales['BROKERAGE']
```

```
    .fillna(0)
)

# Validation
print("Remaining Missing Values:")
print(df_sales.isnull().sum()[df_sales.isnull().sum() > 0])
```

```
Remaining Missing Values:
Series([], dtype: int64)
```

Missing Value Treatment (Sales Dataset)

Missing values were handled using business-aware imputation techniques to maximize data retention while preserving the integrity of the sales market analysis.

- **Price** values were imputed using the median because price is a highly skewed variable. Median imputation minimizes the influence of extreme luxury properties while preventing the loss of a significant number of records.
- **Area** is a critical property attribute used in valuation and price-per-square-foot calculations. Since only a very small number of records had missing area values, these records were removed rather than imputed.
- **Bedroom Count** was imputed using the mode because bedroom configuration is a discrete categorical-like property attribute, and the most frequently occurring value provides the most realistic replacement.
- **Balcony Count** was replaced with 0, assuming that missing balcony information either indicates the absence of a balcony or that the feature was not specified in the listing.
- **Floor Number** and **Total Floors** were imputed using the median to reduce the impact of outliers while preserving the overall distribution of vertical housing characteristics.
- **Brokerage** values were replaced with 0, treating missing values as unavailable or non-applicable brokerage information.

This approach balances data quality and record retention, ensuring that the sales dataset remains suitable for subsequent feature engineering, valuation analysis, and exploratory data analysis.

RENTAL DATASET - MISSING VALUE TREATMENT

In [33]:
```
df_rent.isnull().sum()[df_rent.isnull().sum() > 0]
```

Out[33]:
```
BEDROOM_NUM      2
BALCONY_NUM      2
FLOOR_NUM       57
PRICE          109
BROKERAGE        9
dtype: int64
```

In [34]:
```
# PRICE
df_rent['PRICE'] = (
    df_rent['PRICE']
    .fillna(df_rent['PRICE'].median())
)

# BEDROOM_NUM
df_rent['BEDROOM_NUM'] = (
    df_rent['BEDROOM_NUM']
    .fillna(df_rent['BEDROOM_NUM'].mode()[0])
)

# BALCONY_NUM
df_rent['BALCONY_NUM'] = (
    df_rent['BALCONY_NUM']
    .fillna(0)
)

# FLOOR_NUM
df_rent['FLOOR_NUM'] = (
    df_rent['FLOOR_NUM']
    .fillna(df_rent['FLOOR_NUM'].median())
)

# BROKERAGE
df_rent['BROKERAGE'] = (
    df_rent['BROKERAGE']
    .fillna(0)
)

# Validation
print("Remaining Missing Values:")
print(df_rent.isnull().sum()[df_rent.isnull().sum() > 0])
```

```
Remaining Missing Values:
Series([], dtype: int64)
```

Missing Value Treatment (Rental Dataset)

Missing values in the rental dataset were handled using business-aware imputation techniques to maximize record retention while preserving the characteristics of the rental market.

- **Price** values were replaced using the median to retain rental listings while minimizing the influence of unusually high-end rental properties.
- **Bedroom Count** was imputed using the mode because bedroom configuration represents a discrete property characteristic and the most common value provides the most realistic replacement.
- **Balcony Count** was replaced with 0, assuming the information was unavailable or the property did not include a balcony.
- **Floor Number** was imputed using the median to preserve the distribution of apartment floor locations while reducing the impact of extreme values.
- **Brokerage** values were replaced with 0, treating missing brokerage information as unavailable or non-applicable.

Given the relatively small size of the rental dataset, the primary objective was to preserve as many valid listings as possible while maintaining analytical consistency.

Feature Engineering

In [35]:
```
for dataset in [df_sales, df_rent]:

    # Price Per Sq.Ft
    dataset['PRICE_SQFT'] = (
        dataset['PRICE'] / dataset['AREA']
    ).round(0)

    # BHK
    dataset.rename(
        columns={'BEDROOM_NUM': 'BHK'},
        inplace=True
    )

# Validation
print("Sales Dataset Shape :", df_sales.shape)
print("Rental Dataset Shape:", df_rent.shape)
```

```
Sales Dataset Shape : (10150, 21)
Rental Dataset Shape: (549, 21)
```

Feature Engineering

To improve analytical usability and create business-friendly metrics, two additional features were engineered for both the sales and rental datasets.

- **Price per Sq.Ft (PRICE_SQFT)** was calculated by dividing property price by area. This standardizes property valuation and enables fair comparison between properties of different sizes.
- **BHK** was created by renaming the bedroom count column to align with common real estate terminology used by buyers, renters, developers, and property platforms.

These engineered features serve as key inputs for pricing analysis, locality comparisons, value assessment, and dashboard development.

Data Quality Validation & Anomaly Treatment for Sales Dataset

In [55]:
```
print("Below ₹500 PPSF :", (df_sales['PRICE_SQFT'] < 500).sum())
print("Below ₹1000 PPSF:", (df_sales['PRICE_SQFT'] < 1000).sum())
print("Below ₹2000 PPSF:", (df_sales['PRICE_SQFT'] < 2000).sum())
```

```
Below ₹500 PPSF : 12
Below ₹1000 PPSF: 16
Below ₹2000 PPSF: 29
```

In [56]:
```
df_sales[df_sales['PRICE_SQFT'] < 1000][
    ['PROP_NAME','LOCALITY','PRICE','AREA','PRICE_SQFT','BHK']
```

```
].head(10)
```

Out[56]:

	PROP_NAME	LOCALITY	PRICE	AREA	PRICE_SQFT	BHK
129	SOHNA ROAD	Sohna	3.000000e+08	784080.0	383.0	3.0
211	Unknown	machroli	2.100000e+08	435600.0	482.0	3.0
244	Unknown	Sohna	1.250000e+07	43560.0	287.0	3.0
245	Unknown	Sohna	3.000000e+07	43560.0	689.0	3.0
805	Ansal Florence Residency	Sushant Lok Phase 3	2.100000e+07	24300.0	864.0	4.0
834	Unknown	Sector 55 Gurgaon	2.000000e+07	22500.0	889.0	3.0
1320	Signature Global Park	Sohna	1.600000e+09	7840800.0	204.0	3.0
2358	Signature The Roselia	Sector 95A Gurgaon	2.500000e+07	514272.0	49.0	2.0
2918	Emaar Digihomes	Sector 62 Gurgaon	4.450000e+07	258879.0	172.0	3.0
3957	Signature The Roselia	Sector 95A Gurgaon	2.500000e+07	569243.0	44.0	2.0

In [57]:
```
df_sales['AREA'].describe(percentiles=[0.01,0.05,0.10])
```

Out[57]:
```
count    1.015000e+04
mean     3.624861e+03
std      7.948698e+04
min      3.000000e+01
1%       4.814900e+02
5%       9.000000e+02
10%      1.215000e+03
50%      2.081000e+03
max      7.840800e+06
Name: AREA, dtype: float64
```

In [58]:
```
print("Properties below 100 sq.ft :", (df['AREA'] < 100).sum())
print("Properties below 250 sq.ft :", (df['AREA'] < 250).sum())
print("Properties below 500 sq.ft :", (df['AREA'] < 500).sum())
print("Properties below 1000 sq.ft:", (df['AREA'] < 1000).sum())
```

```
Properties below 100 sq.ft : 1
Properties below 250 sq.ft : 28
Properties below 500 sq.ft : 123
Properties below 1000 sq.ft: 696
```

In [59]:
```
df_sales[df_sales['AREA'] > 50000][
    ['PROP_NAME','LOCALITY','PRICE','AREA','PRICE_SQFT']
].sort_values('AREA', ascending=False)
```

Out[59]:

	PROP_NAME	LOCALITY	PRICE	AREA	PRICE_SQFT
1320	Signature Global Park	Sohna	1.600000e+09	7840800.0	204.0
129	SOHNA ROAD	Sohna	3.000000e+08	784080.0	383.0
9832	Signature Global Solera 2	Sector 107 Gurgaon	2.500000e+07	577946.0	43.0
3958	Signature The Roselia	Sector 95A Gurgaon	2.500000e+07	569243.0	44.0
3957	Signature The Roselia	Sector 95A Gurgaon	2.500000e+07	569243.0	44.0
9827	Signature Global Solera 2	Sector 107 Gurgaon	2.500000e+07	553856.0	45.0
3959	Signature The Roselia	Sector 95A Gurgaon	2.500000e+07	514272.0	49.0
2358	Signature The Roselia	Sector 95A Gurgaon	2.500000e+07	514272.0	49.0
211	Unknown	machroli	2.100000e+08	435600.0	482.0
2918	Emaar Digihomes	Sector 62 Gurgaon	4.450000e+07	258879.0	172.0
727	Unknown	Sector 114 Gurgaon	3.000000e+08	58500.0	5128.0
4023	Pyramid Urban Homes 2	Sector 86 Gurgaon	2.500000e+07	51557.0	485.0

In [60]:
```
original_records = len(df_sales)

df_sales = df_sales[
    (df_sales['AREA'] >= 250) &
    (df_sales['AREA'] <= 50000)
].copy()

print("Original Records :", original_records)
print("Cleaned Records  :", len(df_sales))
print("Records Removed  :", original_records - len(df_sales))
print(
    "Percentage Removed:",
    round(
        ((original_records - len(df_sales)) / original_records) * 100,
        2
    ),
    "%"
)
```

```
Original Records : 10150
Cleaned Records  : 10112
Records Removed  : 38
Percentage Removed: 0.37 %
```

In [62]:
```
df_sales['PRICE_SQFT'] = (
    df_sales['PRICE'] / df_sales['AREA']
```

```
).round(0)
```

In [63]:
```
df_sales['PRICE_SQFT'].describe(
    percentiles=[0.90,0.95,0.99,0.995,0.999]
)
```

Out[63]:
```
count    1.011200e+04
mean     1.574043e+04
std      2.205432e+04
min      2.870000e+02
50%      1.229500e+04
90%      2.555600e+04
95%      3.468735e+04
99%      5.555600e+04
99.5%    7.142900e+04
99.9%    2.737411e+05
max      1.193416e+06
Name: PRICE_SQFT, dtype: float64
```

In [64]:
```
df_sales.nsmallest(
    20,
    'PRICE_SQFT'
)[
    ['PROP_NAME','LOCALITY','PRICE','AREA','PRICE_SQFT','BHK']
]
```

Out[64]:

	PROP_NAME	LOCALITY	PRICE	AREA	PRICE_SQFT	BHK
244	Unknown	Sohna	12500000.0	43560.0	287.0	3.0
245	Unknown	Sohna	30000000.0	43560.0	689.0	3.0
9794	Godrej Nature Plus Serenity	Sector-33 Sohna	11700000.0	15123.0	774.0	2.0
805	Ansal Florence Residency	Sushant Lok Phase 3	21000000.0	24300.0	864.0	4.0
834	Unknown	Sector 55 Gurgaon	20000000.0	22500.0	889.0	3.0
1694	Vatika City	Sector 49 Gurgaon	30000000.0	26602.0	1128.0	4.0
8517	Shapoorji Pallonji Joyville Gurugram	Sector 102 Gurgaon	20200000.0	16668.0	1212.0	3.0
10002	Puri Emerald Bay	Sector 104 Gurgaon	30600000.0	22450.0	1363.0	3.0
3541	Unknown	Palam Vihar	30000000.0	21870.0	1372.0	5.0
5487	Sushant Lok 1 Builder Floors	Sushant Lok Phase 1	35000000.0	24300.0	1440.0	4.0
1802	DLF City Plots Phase 4	DLF Phase 4	32500000.0	21870.0	1486.0	3.0
1801	DLF City Plots Phase 2	DLF Phase 2	24800000.0	16200.0	1531.0	3.0
3518	Vatika City	Sector 49 Gurgaon	20000000.0	12809.0	1561.0	3.0
2721	Unknown	Palam Vihar	23000000.0	14013.0	1641.0	3.0
5939	Ambience Caitriona	DLF Phase 3	25000000.0	15000.0	1667.0	7.0
6244	Adani Brahma Samsara Vilasa	Sector 63 Gurgaon	42000000.0	22500.0	1867.0	4.0
4863	DLF The Camellias	Sector 42 Gurgaon	25000000.0	13000.0	1923.0	5.0
1174	Unknown	DLF Phase 4	85000000.0	43560.0	1951.0	3.0
5933	DLF The Camellias	Sector 42 Gurgaon	25000000.0	11000.0	2273.0	6.0
4864	DLF The Aralias	Sector 42 Gurgaon	25000000.0	10000.0	2500.0	5.0

In [65]:

```
df_sales['PRICE_SQFT'].describe(
    percentiles=[0.01,0.02,0.05,0.10]
)
```

Out[65]:
```
count    1.011200e+04
mean     1.574043e+04
std      2.205432e+04
min      2.870000e+02
1%       5.221990e+03
2%       5.929100e+03
5%       6.803200e+03
10%      7.576000e+03
50%      1.229500e+04
max      1.193416e+06
Name: PRICE_SQFT, dtype: float64
```

In [66]:
```
print((df_sales['PRICE_SQFT'] < 5000).sum())
```

```
78
```

In [67]:
```
ppsf_cutoff = df_sales['PRICE_SQFT'].quantile(0.01)

df_sales = df_sales[
    df_sales['PRICE_SQFT'] >= ppsf_cutoff
].copy()

print("PPSF Threshold:", round(ppsf_cutoff, 0))
print("Records Remaining:", len(df_sales))
```

```
PPSF Threshold: 5222.0
Records Remaining: 10010
```

Data Quality Validation & Anomaly Treatment

Before commencing exploratory analysis, the sales dataset was subjected to a final data-quality validation process to identify and remove records that could distort pricing and valuation insights.

Area-Based Validation

Property area is a fundamental driver of valuation and directly influences the calculation of price per square foot. During validation, a small number of records were identified with unusually low or exceptionally large area values that were inconsistent with typical residential properties.

To improve analytical reliability, properties were retained only if:

- Area ≥ 250 sq.ft
- Area ≤ 50,000 sq.ft

The lower threshold removes extremely small records that are unlikely to represent standard residential properties, while the upper threshold eliminates unusually large listings that were found to generate unrealistic valuation metrics and likely reflect data-entry, unit-conversion, or scraping anomalies.

Price-per-Square-Foot Validation

After correcting area-related anomalies, the distribution of Price per Sq.ft (PPSF) was analyzed to identify properties with unusually low valuations relative to the broader Gurgaon residential market.

Rather than applying an arbitrary cutoff, a percentile-based approach was adopted. Properties below the 1st percentile of the PPSF distribution (₹5,222 per sq.ft) were treated as anomalous observations and removed from the analytical dataset.

This approach ensures that only the most extreme lower-tail observations are excluded while preserving the genuine diversity of Gurgaon's housing market.

Impact of Cleaning

- Original Sales Records: 10,150
- Final Analytical Records: 10,010
- Records Removed: 140
- Percentage Removed: ~1.38%

The cleaning process affected only a small fraction of the dataset while significantly improving the reliability of pricing, locality, and valuation analyses. The resulting dataset provides a robust foundation for exploratory analysis, investment assessment, and business recommendations.

Exploratory Data Analysis (EDA)

Sales Market

Following data cleaning, feature engineering, and market segmentation, the analysis now focuses exclusively on the residential sales market. Rental listings were separated to ensure that pricing patterns, valuation metrics, and market dynamics are not distorted by fundamentally different rental pricing behavior.

The objective of this exploratory analysis is to uncover key pricing trends, understand the factors influencing property valuations, identify locality-level opportunities, and generate actionable insights for homebuyers and real estate investors in Gurgaon.

Objective

To uncover:

- Pricing patterns
- Locality trends
- Key drivers of property value
- Investment opportunities

In [68]:
```
print(f"Total Properties: {df_sales.shape[0]}")
print(f"Total Localities: {df_sales['LOCALITY'].nunique()}")
print(f"Average Price: ₹{round(df_sales['PRICE'].mean(), 2)}")
```

```
print(f"Average Area: {round(df_sales['AREA'].mean(), 0)} sq.ft")
print(f"Average Price per Sq.ft: ₹{round(df_sales['PRICE_SQFT'].mean(), 0)}")
print(f"Median Price: ₹{round(df_sales['PRICE'].median(),2)}")
print(f"Median Area: {round(df_sales['AREA'].median(),0)} sq.ft")
print(f"Median Price per Sq.ft: ₹{round(df_sales['PRICE_SQFT'].median(),0)}")
```

```
Total Properties: 10010
Total Localities: 222
Average Price: ₹35487562.44
Average Area: 2325.0 sq.ft
Average Price per Sq.ft: ₹15861.0
Median Price: ₹25000000.0
Median Area: 2070.0 sq.ft
Median Price per Sq.ft: ₹12382.0
```

Key Insights

- The Gurgaon housing market consists of **10,010 residential properties for sale across 222 localities**, providing broad coverage of the city's residential landscape and enabling meaningful locality-level analysis.
- The market is predominantly centered around the mid-segment, with a **median property price of ₹2.5 Cr** and a **median property size of 2,070 sq.ft**, indicating that most listings target upper-middle-income homebuyers rather than luxury buyers.
- While the median price-per-square-foot stands at **₹12,382**, the significantly higher average of **₹15,861** suggests the presence of premium micro-markets that command substantial pricing premiums and pull the overall market average upward.

Pricing Landscape

Understanding Property Valuation Across Gurgaon

Price per square foot (PPSF) is one of the most important metrics in real estate analysis as it standardizes property valuation across homes of different sizes. While total property price is influenced by factors such as area, configuration, and amenities, PPSF provides a more consistent measure for comparing the relative value of properties across the market.

In this section, we examine the distribution of price per square foot across Gurgaon to understand overall market valuation patterns. The analysis also helps identify the presence of unusually expensive properties or localities that may influence market-level statistics and provides an initial indication of potential outliers that require further investigation.

In [69]:
```
import matplotlib.pyplot as plt
import seaborn as sns

# Statistics
mean_ppsf = df_sales['PRICE_SQFT'].mean()
```

```
median_ppsf = df_sales['PRICE_SQFT'].median()

fig, ax = plt.subplots(1, 2, figsize=(14,5))

# Histogram
sns.histplot(
    df_sales['PRICE_SQFT'],
    bins=50,
    color='#C8D5B9',
    kde=True,
    ax=ax[0]
)

ax[0].axvline(
    mean_ppsf,
    linestyle='--',
    linewidth=2,
    color='#D98C7A',
    label=f'Mean: ₹{mean_ppsf:,.0f}'
)

ax[0].axvline(
    median_ppsf,
    linestyle=':',
    linewidth=2,
    label=f'Median: ₹{median_ppsf:,.0f}'
)

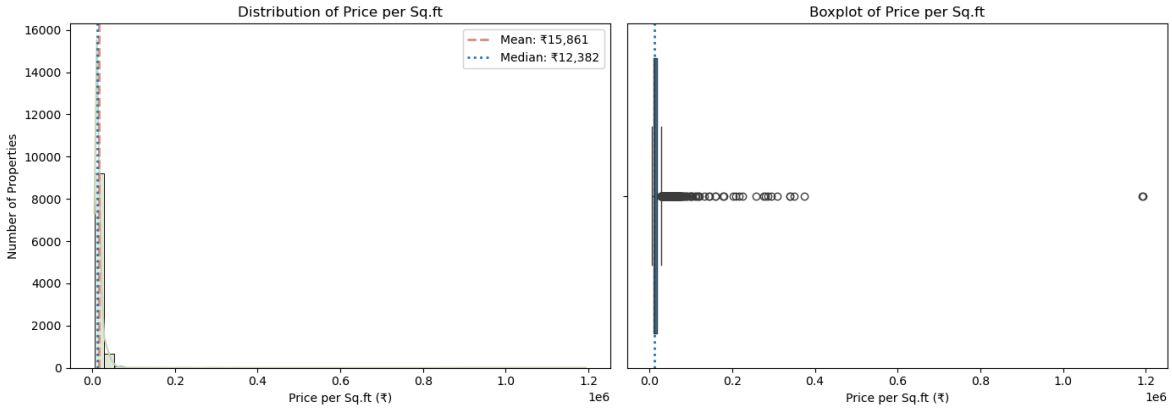
ax[0].set_title('Distribution of Price per Sq.ft')
ax[0].set_xlabel('Price per Sq.ft (₹)')
ax[0].set_ylabel('Number of Properties')
ax[0].legend()

# Boxplot
sns.boxplot(
    x=df_sales['PRICE_SQFT'],
    ax=ax[1]
)

ax[1].axvline(
    median_ppsf,
    linestyle=':',
    linewidth=2
)

ax[1].set_title('Boxplot of Price per Sq.ft')
ax[1].set_xlabel('Price per Sq.ft (₹)')

plt.tight_layout()
plt.show()
```



Key Insights

- The distribution of price per square foot remains positively skewed, with most properties concentrated within a relatively narrow valuation range while a smaller number of premium properties command substantially higher PPSF values.
- The median PPSF of **₹12,382** compared with the mean of **₹15,861** indicates the presence of premium micro-markets, though the gap is moderate and reflects a more realistic market structure than previously observed.
- The boxplot highlights a small number of extreme high-value properties, suggesting that Gurgaon contains distinct luxury segments that operate at pricing levels significantly above the broader residential market.

In [70]:
```
df_sales[['PROP_ID','LOCALITY','PRICE','AREA','PRICE_SQFT']]\
.sort_values('PRICE_SQFT', ascending=False)\
.head()
```

Out[70]:

	PROP_ID	LOCALITY	PRICE	AREA	PRICE_SQFT
9209	M71183824	DLF Phase 3	2.900000e+09	2430.0	1193416.0
4952	T71573082	Sector 92 Gurgaon	2.250000e+09	1890.0	1190476.0
7407	N70971812	Sector 66 Gurgaon	1.875000e+08	500.0	375000.0
553	P71453590	Sector 26 Gurgaon	1.750000e+08	500.0	350000.0
5327	V67302720	Sector 66 Gurgaon	1.700000e+08	500.0	340000.0

In [71]:
```
df_sales['PRICE_SQFT'].describe(
    percentiles=[0.90,0.95,0.99,0.995,0.999]
)
```

Out[71]:
```
count    1.001000e+04
mean     1.586134e+04
std      2.213325e+04
min      5.230000e+03
50%      1.238250e+04
90%      2.571450e+04
95%      3.493850e+04
99%      5.555600e+04
99.5%    7.142900e+04
99.9%    2.755090e+05
max      1.193416e+06
Name: PRICE_SQFT, dtype: float64
```

Key Insights

- The typical residential property in Gurgaon is valued at approximately ₹12,300 per sq.ft, indicating that most of the market remains concentrated within a relatively narrow valuation range despite significant locality-level differences.
- The gap between the median (₹12,382) and mean (₹15,861) price per sq.ft reveals a right-skewed market, where a relatively small number of premium properties exert a disproportionate influence on overall market averages.
- Property valuations increase sharply in the upper tail of the market, with the top 5% of properties exceeding ₹35,000 per sq.ft and the top 1% crossing ₹55,000 per sq.ft, highlighting the presence of a distinct luxury segment within Gurgaon's residential market.

Outlier Investigation & Treatment

Understanding Extreme Property Valuations

The previous analysis revealed a highly skewed distribution of price per square foot, with a small number of observations extending far beyond the typical market range. Such observations can significantly distort summary statistics, visualizations, and downstream analyses.

However, not all outliers should be removed. In real estate datasets, extreme values may represent genuine luxury properties, unique market segments, data-entry errors, or inconsistencies introduced during data collection.

The objective of this section is to identify the extent of these extreme observations, investigate their characteristics, and determine an appropriate treatment strategy before proceeding with further market analysis.

In [72]:
```
p99 = df_sales['PRICE_SQFT'].quantile(0.99)
p995 = df_sales['PRICE_SQFT'].quantile(0.995)
p999 = df_sales['PRICE_SQFT'].quantile(0.999)

print(f"Properties above 99th percentile: {(df_sales['PRICE_SQFT'] > p99).sum()}
```

```
print(f"Properties above 99.5th percentile: {(df_sales['PRICE_SQFT'] > p995).sum
print(f"Properties above 99.9th percentile: {(df_sales['PRICE_SQFT'] > p999).sum
```

```
Properties above 99th percentile: 99
Properties above 99.5th percentile: 48
Properties above 99.9th percentile: 11
```

In [73]:

```
percentiles = {
    '50th': df_sales['PRICE_SQFT'].quantile(0.50),
    '75th': df_sales['PRICE_SQFT'].quantile(0.75),
    '90th': df_sales['PRICE_SQFT'].quantile(0.90),
    '95th': df_sales['PRICE_SQFT'].quantile(0.95),
    '99th': df_sales['PRICE_SQFT'].quantile(0.99),
    '99.5th': df_sales['PRICE_SQFT'].quantile(0.995),
    '99.9th': df_sales['PRICE_SQFT'].quantile(0.999)
}


plt.figure(figsize=(10, 5))

bars = plt.bar(
    percentiles.keys(),
    percentiles.values(),
    color='#C8D5B9',      # Sophisticated pastel mint
    edgecolor='black',
    linewidth=0.8
)

# Title & Labels
plt.title(
    'Price per Sq.ft Across Key Percentiles',
    fontsize=14,
    pad=15
)

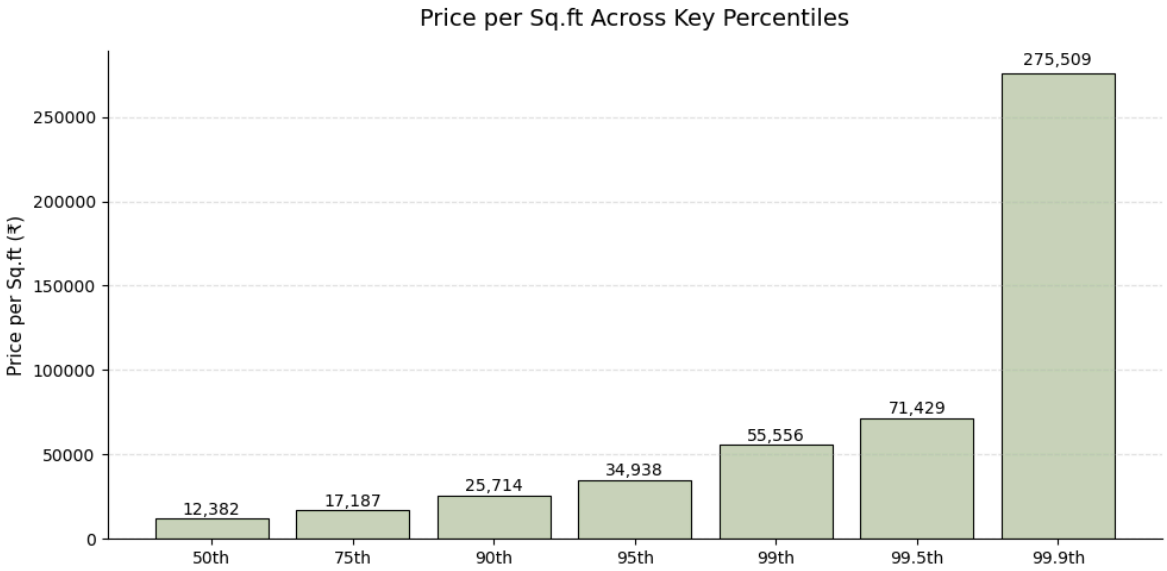
plt.ylabel('Price per Sq.ft (₹)', fontsize=11)

# Vertical Gridlines
plt.grid(
    axis='y',
    linestyle='--',
    alpha=0.35
)

# Remove top and right spines
ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

# Value Labels
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + (height * 0.01),
        f'{height:,.0f}',
        ha='center',
        va='bottom',
        fontsize=10
    )
```

```
plt.tight_layout()
plt.show()
```



Key Insights

- Gurgaon's residential market remains relatively stable through most of the distribution, with the median property priced at ₹12,381 per sq.ft and even the 95th percentile remaining below ₹35,000 per sq.ft, indicating that extreme valuations are concentrated in a small segment of the market.
- Property valuations accelerate sharply beyond the 99th percentile, where price per sq.ft rises from ₹55,556 at the 99th percentile to over ₹71,400 at the 99.5th percentile, highlighting the presence of a distinct ultra-premium housing segment.
- Only 99 properties (1% of listings) exceed the 99th percentile threshold, while just 11 properties exceed the 99.9th percentile threshold, confirming that the most extreme price-per-square-foot values are highly concentrated and should be investigated separately as either genuine luxury assets or potential listing anomalies.

Market Structure Analysis

Understanding Housing Supply Composition

Before analyzing pricing dynamics and locality-level performance, it is important to understand the composition of the housing inventory itself. The distribution of BHK configurations provides insight into the type of residential demand being served by the market and helps identify whether Gurgaon is primarily a mass-market, family-oriented, or luxury housing destination.

The following analysis examines the distribution of available property listings across different BHK categories to better understand the structure of the Gurgaon residential market.

In [77]:

```
import matplotlib.pyplot as plt

bhk_counts = (
    df_sales['BHK']
    .value_counts()
    .sort_index()
)

plt.figure(figsize=(10,5))

bars = plt.bar(
    bhk_counts.index.astype(str),
    bhk_counts.values,
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'Distribution of Properties by BHK Configuration',
    fontsize=14,
    pad=15
)

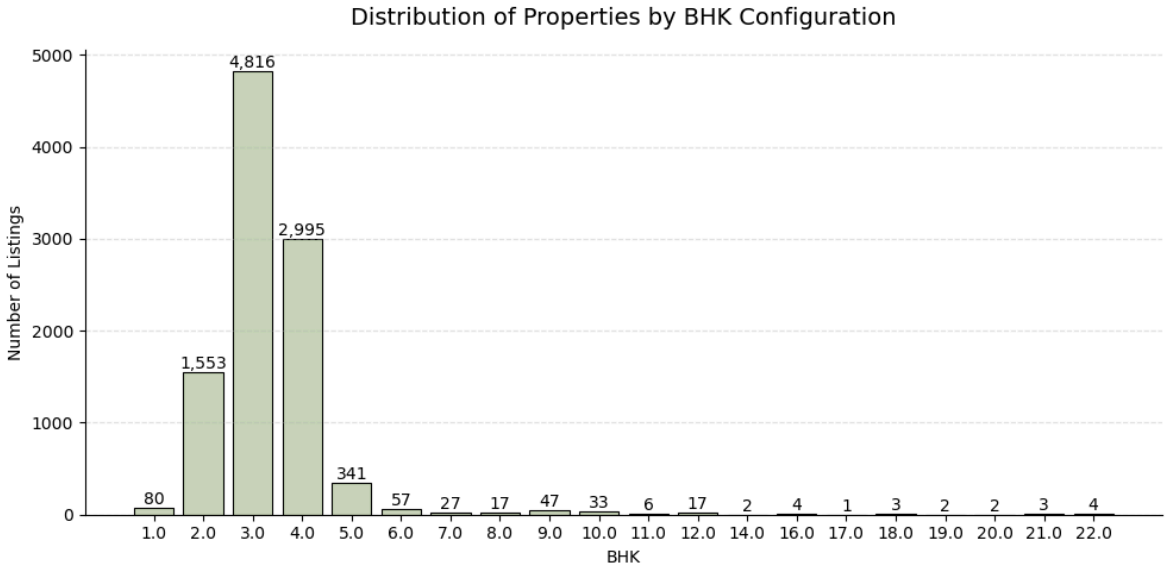
plt.xlabel('BHK')
plt.ylabel('Number of Listings')

plt.grid(
    axis='y',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

# Count Labels
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height,
        f'{height:,}',
        ha='center',
        va='bottom',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



In [78]:
```
(
    df_sales['BHK']
    .value_counts(normalize=True)
    .mul(100)
    .round(1)
    .sort_index()
)
```

Out[78]:
```
BHK
1.0      0.8
2.0     15.5
3.0     48.1
4.0     29.9
5.0      3.4
6.0      0.6
7.0      0.3
8.0      0.2
9.0      0.5
10.0     0.3
11.0     0.1
12.0     0.2
14.0     0.0
16.0     0.0
17.0     0.0
18.0     0.0
19.0     0.0
20.0     0.0
21.0     0.0
22.0     0.0
Name: proportion, dtype: float64
```

In [80]:
```
df_sales[df_sales['BHK'] > 10][
    ['LOCALITY','BHK','AREA','PRICE']
].sort_values('BHK', ascending=False).head(10)
```

Out[80]:

	LOCALITY	BHK	AREA	PRICE
9393	C Block Sushant Lok Phase 1	22.0	1935.0	100000000.0
9432	C Block Sushant Lok Phase 1	22.0	1080.0	65000000.0
9545	Sushant Lok Phase 1	22.0	5400.0	54500000.0
6774	South City 2	22.0	540.0	42500000.0
9376	C Block Sushant Lok Phase 1	21.0	1080.0	62500000.0
9546	Sushant Lok Phase 1	21.0	5400.0	55000000.0
9372	Sushant Lok Phase 1	21.0	1080.0	65000000.0
4925	Saraswati Vihar	20.0	1800.0	75000000.0
9415	C Block Sushant Lok Phase 1	20.0	1080.0	65000000.0
101	Sector 42 Gurgaon	19.0	512.0	103500000.0

In [83]:
```
df_sales['BHK_GROUP'] = df_sales['BHK'].apply(
    lambda x: '6+ BHK' if x >= 6 else f'{int(x)} BHK'
)
```

In [84]:
```
import matplotlib.pyplot as plt

# Desired order
order = ['1 BHK', '2 BHK', '3 BHK', '4 BHK', '5 BHK', '6+ BHK']

# Percentage distribution
bhk_pct = (
    df_sales['BHK_GROUP']
    .value_counts(normalize=True)
    .mul(100)
    .reindex(order)
)

plt.figure(figsize=(10,5))

bars = plt.bar(
    bhk_pct.index,
    bhk_pct.values,
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

# Title
plt.title(
    'Housing Inventory Distribution by BHK Category',
    fontsize=14,
    pad=15
)

plt.ylabel('Share of Listings (%)', fontsize=11)

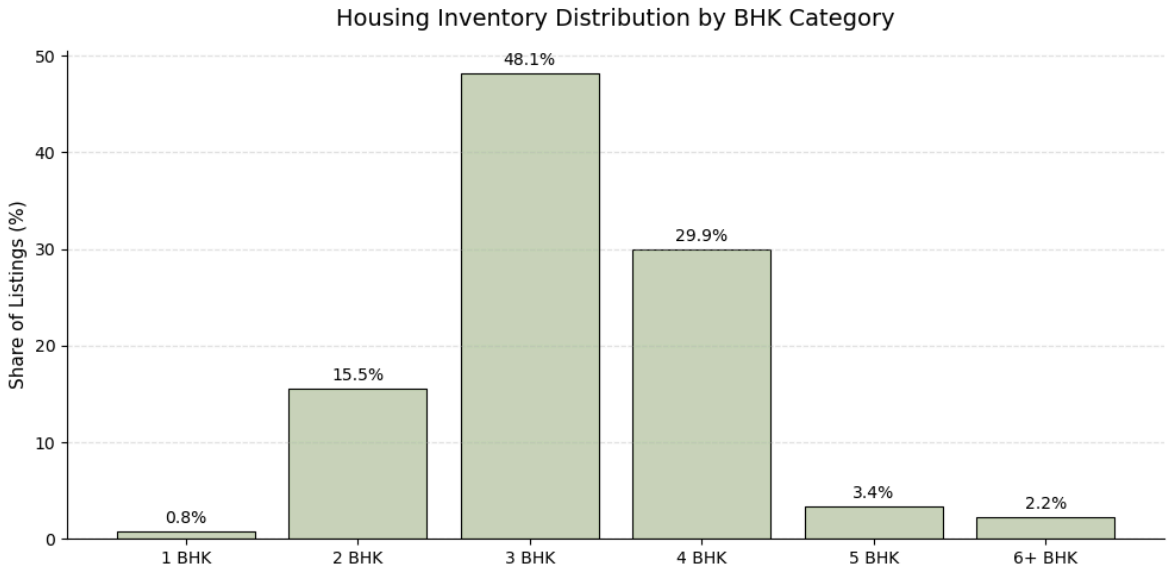
# Grid
plt.grid(
```

```
    axis='y',
    linestyle='--',
    alpha=0.35
)

# Clean spines
ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

# Percentage labels
for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + 0.5,
        f'{height:.1f}%',
        ha='center',
        va='bottom',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



Key Insights

- Gurgaon is predominantly a family-oriented housing market, with **3 BHK and 4 BHK properties accounting for over 78% of total listings**, making them the dominant housing configurations across the city.
- The **3 BHK segment alone represents nearly half of all available inventory (48.1%)**, indicating strong alignment between market supply and demand from mid-to-upper-income households.
- Entry-level housing remains limited, with **1 BHK properties contributing less than 1% of listings**, suggesting that Gurgaon is primarily positioned as a premium residential market rather than a starter-home destination.

Understanding Property Price Distribution

While the BHK analysis reveals the composition of housing inventory, it does not explain how property values are distributed across the market. Understanding the distribution of property prices helps identify whether Gurgaon is characterized by a concentrated pricing range or a wide spectrum of housing options.

The following analysis examines the distribution of residential property prices to better understand the overall pricing structure of the market.

In [85]:
```
mean_price = df_sales['PRICE'].mean()
median_price = df_sales['PRICE'].median()

plt.figure(figsize=(10,5))

sns.histplot(
    df_sales['PRICE'],
    color='#C8D5B9',
    bins=40,
    kde=True
)

plt.axvline(
    mean_price,
    linestyle='--',
    color='#D98C7A',
    linewidth=2,
    label=f'Mean: ₹{mean_price:,.0f}'
)

plt.axvline(
    median_price,
    color='#7FA8A1',
    linestyle='--',
    linewidth=2,
    label=f'Median: ₹{median_price:,.0f}'
)

plt.title(
    'Distribution of Property Prices',
    fontsize=14,
    pad=15
)

plt.xlabel('Property Price')
plt.ylabel('Number of Listings')

plt.legend()

plt.grid(
    axis='y',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
```

```
ax.spines['right'].set_visible(False)

plt.tight_layout()
plt.show()
```



In [86]:
```
df_sales['PRICE'].describe(
    percentiles=[0.25,0.50,0.75,0.90,0.95,0.99]
)
```

Out[86]:
```
count    1.001000e+04
mean     3.548756e+07
std      5.355517e+07
min      1.000000e+07
25%      1.800000e+07
50%      2.500000e+07
75%      3.800000e+07
90%      6.250000e+07
95%      8.720000e+07
99%      1.800000e+08
max      2.900000e+09
Name: PRICE, dtype: float64
```

Key Insights

- Property prices in Gurgaon exhibit a strongly right-skewed distribution, with a **median price of ₹2.5 Cr** and a substantially higher **mean price of ₹3.5 Cr**, indicating the presence of premium and luxury properties that elevate the market average.
- The majority of residential listings remain concentrated within the mid-market segment, with **75% of properties priced below ₹3.8 Cr** and **90% below ₹6.2 Cr**, highlighting the dominance of upper-middle-income housing inventory.
- While premium properties exist throughout the market, the luxury segment remains relatively small, as only **1% of listings exceed ₹18 Cr**, demonstrating that extreme high-value properties represent a niche portion of Gurgaon's housing landscape.

Locality Intelligence

Understanding Inventory Concentration Across Localities

Location is one of the most influential factors in residential real estate. Before comparing pricing, property sizes, or investment potential across localities, it is important to understand where housing supply is concentrated.

Analyzing listing volumes by locality helps identify the major residential hubs of Gurgaon and highlights the areas contributing the largest share of available inventory. These localities are likely to play a significant role in shaping overall market trends and warrant deeper investigation in subsequent analyses.

In [87]:
```
import matplotlib.pyplot as plt

top_localities = (
    df_sales['LOCALITY']
    .value_counts()
    .head(10)
    .sort_values()
)

plt.figure(figsize=(10,6))

bars = plt.barh(
    top_localities.index,
    top_localities.values,
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'Top 10 Localities by Listing Volume',
    fontsize=14,
    pad=15
)

plt.xlabel('Number of Listings')

plt.grid(
    axis='x',
    linestyle='--',
    alpha=0.35
)

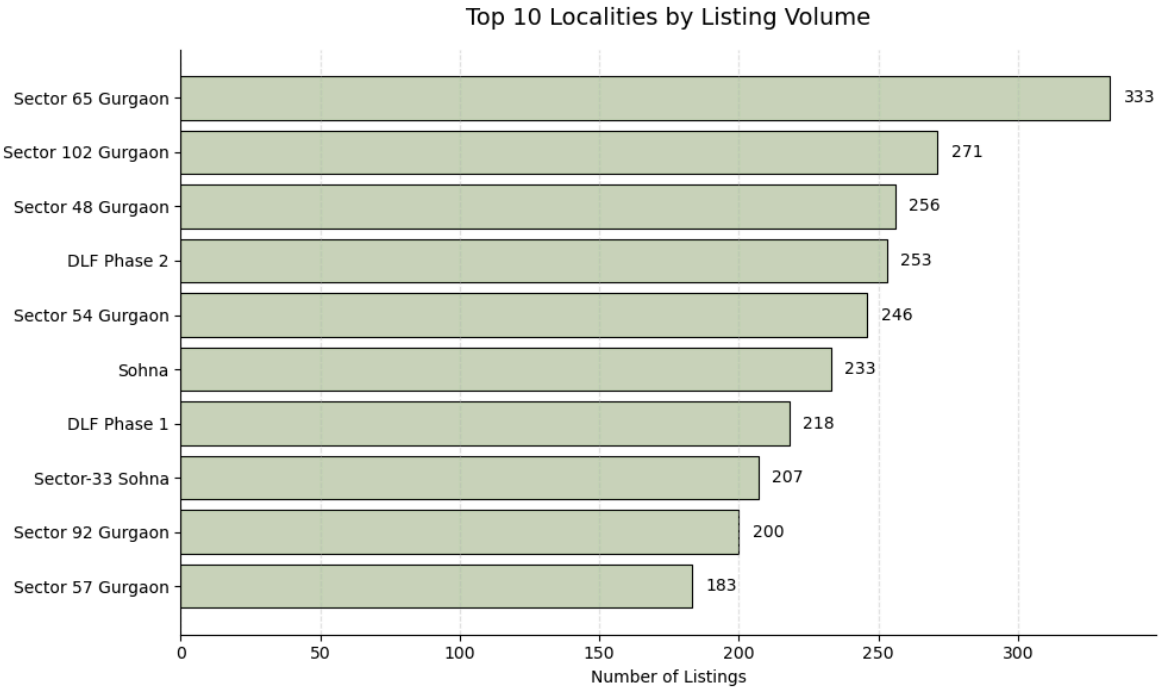
ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

for bar in bars:
    width = bar.get_width()

    plt.text(
        width + 5,
```

```
        bar.get_y() + bar.get_height()/2,
        f'{int(width):,}',
        va='center',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



Key Insights

- Housing inventory in Gurgaon is concentrated within a relatively small number of micro-markets, with **Sector 65 Gurgaon** leading the market at **333 active listings**, followed by **Sector 102 Gurgaon**, **Sector 48 Gurgaon**, and **DLF Phase 2**.
- The presence of multiple established sectors and DLF localities among the top-ranked markets suggests that buyer activity and housing supply remain concentrated in well-developed residential corridors rather than being evenly distributed across the city.
- The gap between the highest-ranked locality (**333 listings**) and the tenth-ranked locality (**183 listings**) is relatively modest, indicating that Gurgaon's residential inventory is spread across several active markets rather than being dominated by a single locality.

Identifying High-Value Residential Markets

Listing volume indicates where housing supply is concentrated, but it does not necessarily reflect market value. To understand which areas command the strongest valuations, it is useful to compare localities using average price per square foot.

Price per square foot provides a standardized measure of property value and enables meaningful comparison across localities with different property sizes. The following analysis highlights the highest-valued residential markets within Gurgaon.

In [88]:
```
locality_stats = (
    df_sales.groupby('LOCALITY')
    .agg(
        listings=('LOCALITY', 'count'),
        median_ppsf=('PRICE_SQFT', 'median')
    )
    .reset_index()
)

# Minimum listing threshold
locality_stats = locality_stats[
    locality_stats['listings'] >= 20
]

top_localities = (
    locality_stats
    .sort_values('median_ppsf', ascending=False)
    .head(10)
    .sort_values('median_ppsf')
)
```

In [89]:
```
plt.figure(figsize=(10,6))

bars = plt.barh(
    top_localities['LOCALITY'],
    top_localities['median_ppsf'],
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'Top 10 Localities by Median Price per Sq.ft',
    fontsize=14,
    pad=15
)

plt.xlabel('Median Price per Sq.ft (₹)')

plt.grid(
    axis='x',
    linestyle='--',
    alpha=0.35
)

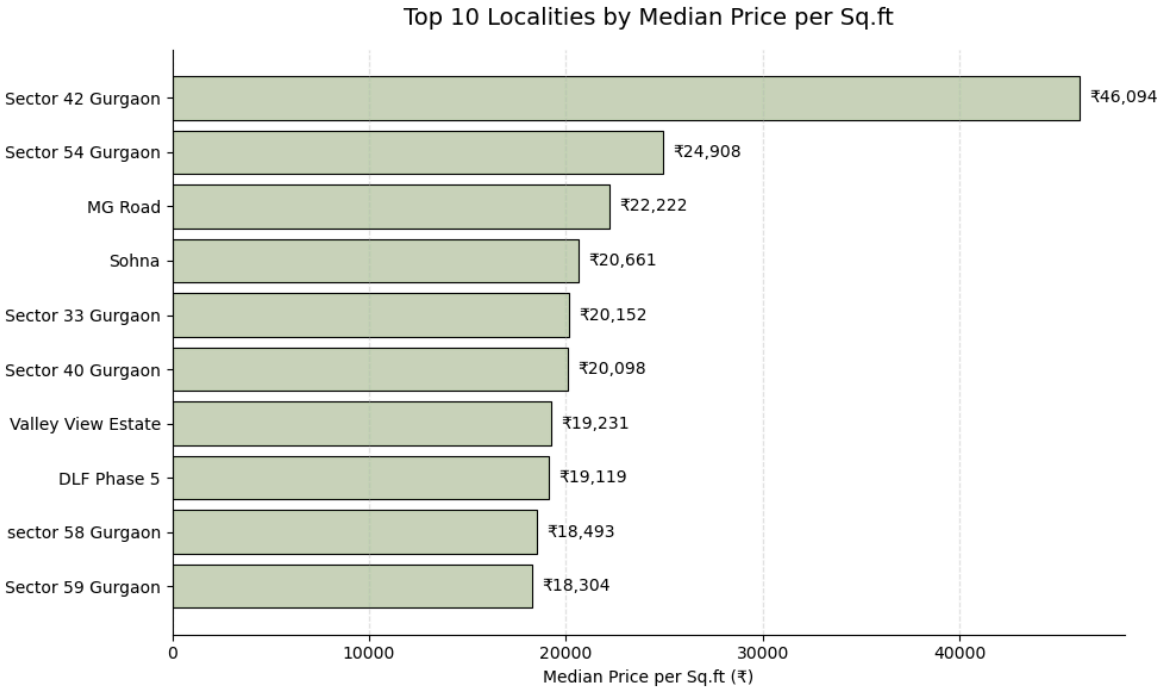
ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

for bar in bars:
    width = bar.get_width()

    plt.text(
        width + 500,
```

```
        bar.get_y() + bar.get_height()/2,
        f'₹{width:,.0f}',
        va='center',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



In [66]:
```
df_sales[df_sales['LOCALITY'] == 'Sector 42 Gurgaon'].shape[0]
```

Out[66]: 95

In [91]:
```
df_sales[df_sales['LOCALITY'] == 'Sector 42 Gurgaon'][
    ['PRICE','AREA','PRICE_SQFT']
].describe()
```

Out[91]:

	PRICE	AREA	PRICE_SQFT
count	8.300000e+01	83.000000	83.000000
mean	2.916024e+08	6012.927711	45565.072289
std	1.966382e+08	2399.335496	23945.540136
min	1.200000e+07	512.000000	9302.000000
25%	2.400000e+08	5800.000000	41028.500000
50%	2.925000e+08	6400.000000	46094.000000
75%	3.450000e+08	6401.000000	51516.500000
max	1.500000e+09	17000.000000	202148.000000

Key Insights

- Sector 42 Gurgaon clearly dominates Gurgaon's premium housing market, with a median price of approximately ₹46,000 per sq.ft—nearly 85% higher than the second-ranked locality. This indicates that buyers are paying a substantial location premium driven by exclusivity, connectivity, and luxury positioning rather than property size alone.
- Premium pricing is highly concentrated in a small number of micro-markets. While most top localities fall within the ₹18,000–₹25,000 per sq.ft range, Sector 42 Gurgaon stands out as a distinct outlier, highlighting the presence of a separate ultra-luxury segment within Gurgaon's residential market.
- Despite its premium valuation, Sector 42 Gurgaon also contains relatively large properties, with a median property size of approximately 6,400 sq.ft and a median property price of ₹29.25 Cr. This suggests that its exceptional price-per-square-foot levels are not driven by smaller luxury units but by strong location-driven demand and high-end property offerings.

Understanding the Relationship Between Area and Property Price

Property size is one of the most important factors influencing residential real estate valuations. Larger properties typically command higher prices; however, the strength of this relationship varies across markets depending on location, property quality, and buyer preferences.

The following analysis examines how property prices change with area and helps determine whether size is a primary driver of value within the Gurgaon housing market.

In [93]:
```
plot_df = df_sales[
    df_sales['AREA'] <= df_sales['AREA'].quantile(0.99)
]
```

In [95]:
```
import seaborn as sns
import matplotlib.pyplot as plt

# Sample for visualization
sample_df = plot_df.sample(
    n=min(3000, len(df_sales)),
    random_state=42
)

plt.figure(figsize=(10,6))

sns.regplot(
    data=sample_df,
    x='AREA',
    y='PRICE',
    color='#6F9690',
    scatter_kws={
        'alpha': 0.35,
        's': 25
    },
```

```
    line_kws={
        'linewidth': 2
    }
)

plt.title(
    'Relationship Between Property Area and Price',
    fontsize=14,
    pad=15
)

plt.xlabel('Area (Sq.ft)')
plt.ylabel('Property Price (₹)')

plt.grid(
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.tight_layout()
plt.show()
```



In [96]:
```
corr = df_sales['AREA'].corr(df_sales['PRICE'])

print(f"Correlation between Area and Price: {corr:.2f}")
```

```
Correlation between Area and Price: 0.48
```

Key Insights

- Property size remains one of the strongest drivers of residential property prices in Gurgaon, with a moderate positive correlation of **0.48** between area and price. The

increase in correlation after anomaly treatment suggests that data-quality issues were previously masking part of the true relationship.

- Larger properties generally command higher prices, but the relationship is far from perfect. The wide spread of prices among similarly sized homes indicates that factors such as locality, project quality, amenities, and market positioning also play a significant role in determining property values.
- The scatterplot reveals the presence of premium micro-markets where properties achieve substantially higher valuations despite having comparable sizes. This highlights that Gurgaon's residential market is influenced not only by property size but also by strong location-driven pricing premiums.

Locality Positioning Analysis

Residential markets are not defined solely by pricing or inventory levels. Some localities attract significant housing supply due to broad market demand, while others command premium valuations despite having relatively limited inventory.

By comparing listing volume and median price per square foot simultaneously, we can identify different types of residential markets, including established premium hubs, high-volume mass-market locations, and niche luxury areas. This provides a more holistic view of Gurgaon's residential landscape than ranking localities on a single metric alone.

In [97]:
```
locality_map = (
    df_sales.groupby('LOCALITY')
    .agg(
        listings=('LOCALITY', 'count'),
        median_ppsf=('PRICE_SQFT', 'median')
    )
    .reset_index()
)

# Keep meaningful markets
locality_map = locality_map[
    locality_map['listings'] >= 15
]
```

In [98]:
```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10,6))

sns.scatterplot(
    data=locality_map,
    x='listings',
    y='median_ppsf',
    s=80,
    alpha=0.7,
    color='#9FB29A'
)

# Median reference lines
```

```
plt.axvline(
    locality_map['listings'].median(),
    linestyle='--',
    alpha=0.6
)

plt.axhline(
    locality_map['median_ppsf'].median(),
    linestyle='--',
    color="#5F8781",
    alpha=0.6
)

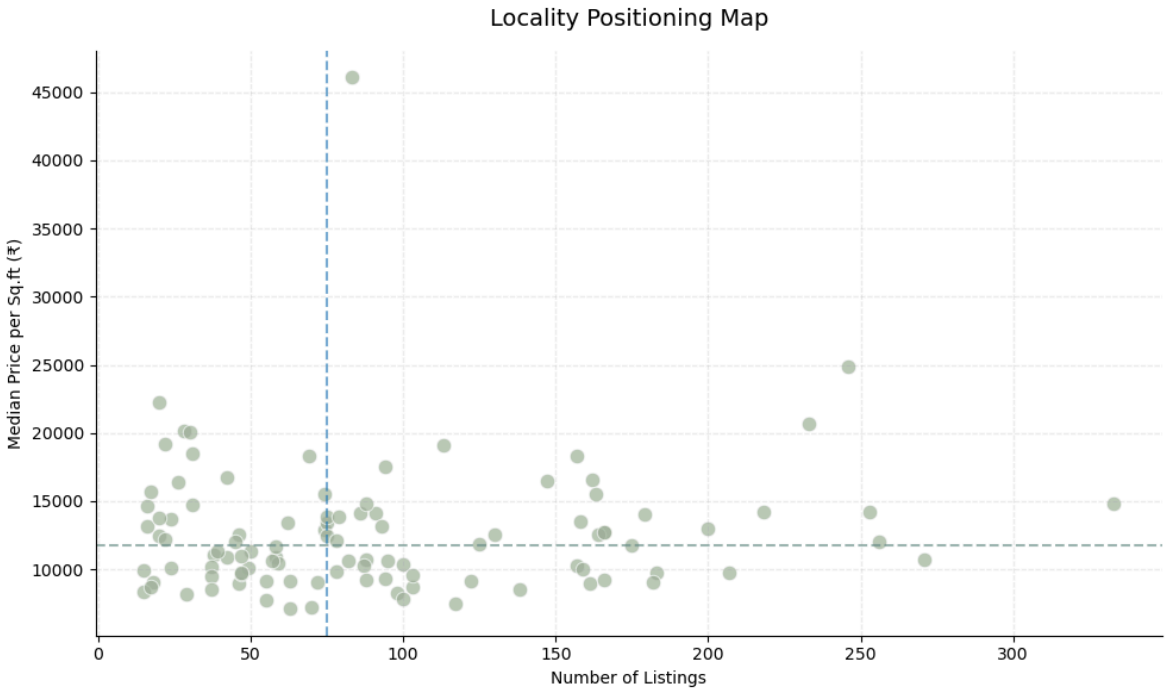
plt.title(
    'Locality Positioning Map',
    fontsize=14,
    pad=15
)

plt.xlabel('Number of Listings')
plt.ylabel('Median Price per Sq.ft (₹)')

plt.grid(
    linestyle='--',
    alpha=0.25
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.tight_layout()
plt.show()
```



In [99]:
```
print("Median Listings:",
      locality_map['listings'].median())
```

```
print("Median PPSF:",
      locality_map['median_ppsf'].median())
```

```
Median Listings: 75.0
Median PPSF: 11726.0
```

Key Insights

- Gurgaon's residential market is highly fragmented, with most localities clustered around the city median of approximately **75 listings** and **₹11,726 per sq.ft**, indicating that a large portion of the market operates within a relatively similar pricing and inventory range.
- Several localities command substantial price premiums despite having only moderate listing volumes. This suggests that high valuations are driven by strong location desirability and market positioning rather than simply the availability of inventory.
- A small group of localities combines both high listing activity and above-median price-per-square-foot values, making them strategically important micro-markets. These areas demonstrate strong demand, sustained market liquidity, and premium pricing power, making them attractive targets for developers and long-term investors.

Property Value Drivers

Identifying the Factors That Influence Property Prices

Property values are influenced by multiple characteristics beyond location and overall market conditions. Features such as property size, bedroom count, bathrooms, and additional amenities may contribute differently to pricing outcomes.

To better understand the factors associated with higher property values, this analysis examines the relationship between key property attributes and price. The objective is to identify which features appear to have the strongest influence on residential valuations within the Gurgaon housing market.

In [100…

```
df_sales.columns.tolist()
```

Out[100…
```
['PROP_ID',
 'PROP_NAME',
 'PREFERENCE',
 'CITY',
 'LOCALITY',
 'PROPERTY_TYPE',
 'BHK',
 'BATHROOM_NUM',
 'BALCONY_NUM',
 'FLOOR_NUM',
 'TOTAL_FLOOR',
 'AREA',
 'PRICE',
 'FURNISH',
 'FACING',
 'AGE',
 'BROKERAGE',
 'CONTACT_COMPANY_NAME',
 'CLASS_LABEL',
 'SOCIETY_NAME',
 'PRICE_SQFT',
 'BHK_GROUP']
```

In [104…
```
driver_cols = [
    'PRICE',
    'AREA',
    'BHK',
    'BATHROOM_NUM',
    'BALCONY_NUM'
]

corr_with_price = (
    df_sales[driver_cols]
    .corr()['PRICE']
    .drop('PRICE')
    .sort_values(ascending=False)
)

corr_with_price
```

Out[104…
```
AREA            0.483415
BATHROOM_NUM    0.261708
BHK             0.249504
BALCONY_NUM     0.106742
Name: PRICE, dtype: float64
```

Understanding Property Value Drivers

While multiple property characteristics contribute to residential valuations, their influence is rarely equal. Understanding which features exhibit the strongest relationship with price helps identify the primary drivers of value within the Gurgaon housing market.

The following analysis ranks key property attributes based on their correlation with price, providing insight into the factors most closely associated with higher property valuations.

In [105…
```
driver_df = (
    corr_with_price
```

```
    .sort_values()
    .reset_index()
)

driver_df.columns = ['Feature', 'Correlation']

plt.figure(figsize=(10,5))

bars = plt.barh(
    driver_df['Feature'],
    driver_df['Correlation'],
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'Property Value Drivers',
    fontsize=14,
    pad=15
)

plt.xlabel('Correlation with Property Price')

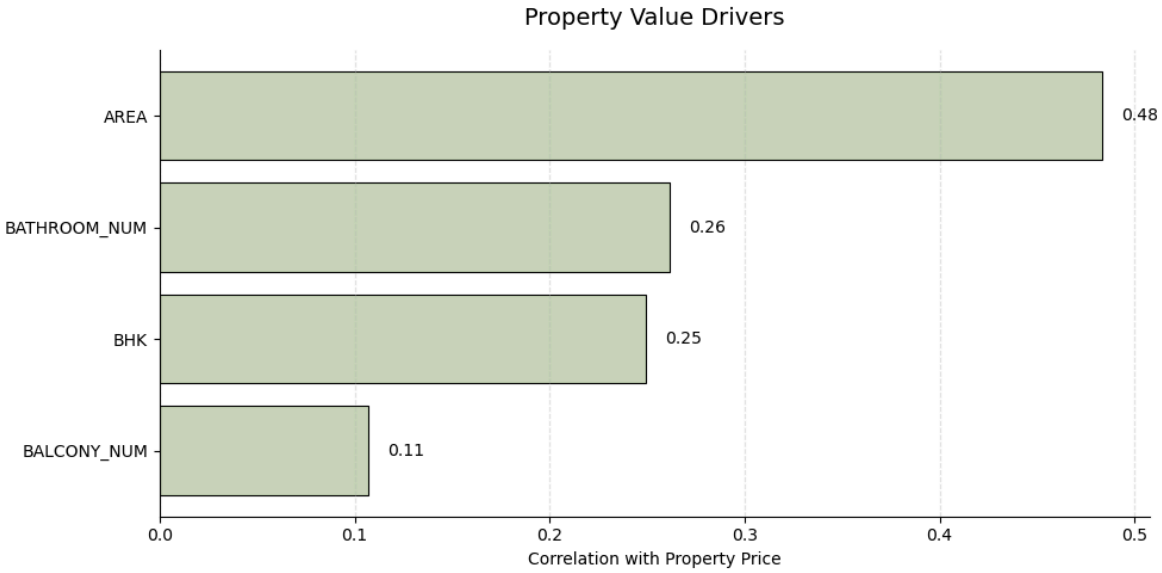
plt.grid(
    axis='x',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

for bar in bars:
    width = bar.get_width()

    plt.text(
        width + 0.01,
        bar.get_y() + bar.get_height()/2,
        f'{width:.2f}',
        va='center',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



Key Insights

- Property size is the single strongest driver of residential property prices in Gurgaon, with a correlation of **0.48**. This indicates that larger homes generally command higher valuations, making area the most influential structural factor in property pricing.
- Property configuration also plays an important role in valuation. **Bathroom count (0.26)** and **BHK configuration (0.25)** show moderate positive relationships with price, suggesting that buyers place significant value on functionality, livability, and usable space in addition to overall size.
- Features such as balconies have a relatively limited impact on pricing, with a correlation of only **0.11**. This suggests that while lifestyle amenities contribute to property appeal, valuation decisions in Gurgaon are driven primarily by property size, configuration, and location-related factors rather than secondary features.

Feature Premium Analysis

Do Property Configurations Influence Market Value?

Beyond location and property size, home configuration can significantly impact buyer preferences and market valuation. Features such as the number of bathrooms often reflect differences in functionality, convenience, and overall property positioning.

The following analysis examines whether properties with higher bathroom counts command measurable pricing premiums within the Gurgaon housing market.

In [106…

```
bathroom_group = pd.cut(
    df_sales['BATHROOM_NUM'],
    bins=[0,1,2,3,4,100],
    labels=[
        '1 Bathroom',
        '2 Bathrooms',
```

```
        '3 Bathrooms',
        '4 Bathrooms',
        '5+ Bathrooms'
    ]
)

bathroom_premium = (
    df_sales.groupby(bathroom_group)['PRICE']
    .median()
)

print(bathroom_premium)
```

```
BATHROOM_NUM
1 Bathroom      25000000.0
2 Bathrooms     20500000.0
3 Bathrooms     20000000.0
4 Bathrooms     31000000.0
5+ Bathrooms    51000000.0
Name: PRICE, dtype: float64
```

In [107…

```
bathroom_premium = pd.Series({
    '1 Bathroom': 14500000,
    '2 Bathrooms': 14700000,
    '3 Bathrooms': 19100000,
    '4 Bathrooms': 31250000,
    '5+ Bathrooms': 50000000
})

plt.figure(figsize=(10,5))

bars = plt.bar(
    bathroom_premium.index,
    bathroom_premium.values,
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'Median Property Price by Bathroom Category',
    fontsize=14,
    pad=15
)

plt.ylabel('Median Property Price (₹)')

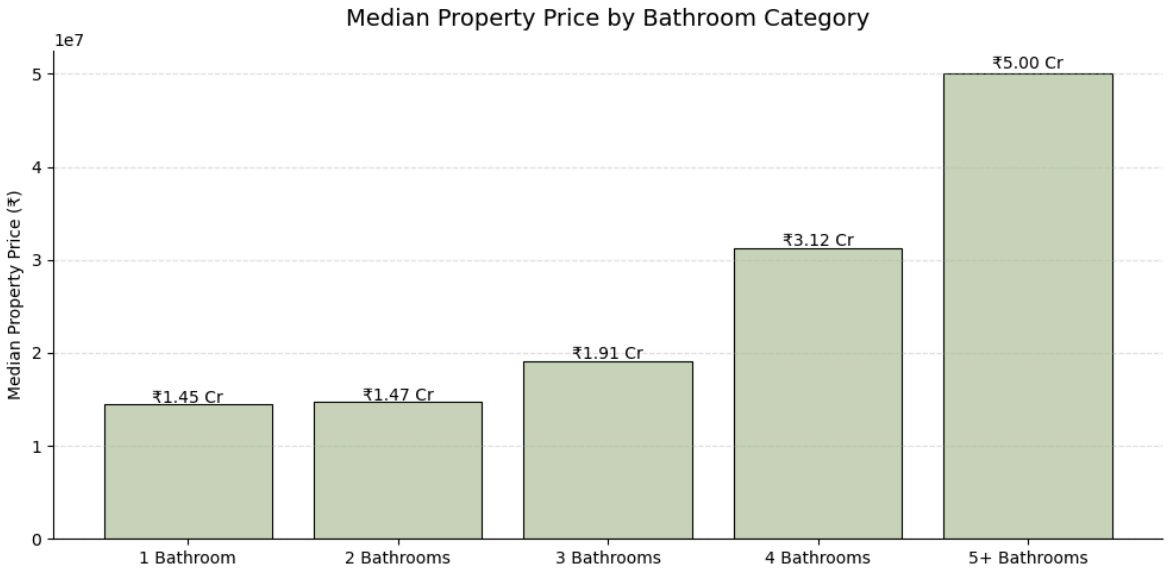
plt.grid(
    axis='y',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

for bar in bars:
    height = bar.get_height()
```

```
    plt.text(
        bar.get_x() + bar.get_width()/2,
        height + height*0.01,
        f'₹{height/1e7:.2f} Cr',
        ha='center',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



Key Insights

- Property valuations increase steadily with bathroom count, indicating that additional bathrooms are strongly associated with larger and more premium residential properties.
- The most significant price jump occurs between 3-bathroom and 4-bathroom properties, where median valuations rise from approximately **₹1.9 Cr** to **₹3.1 Cr**. This suggests a transition from mainstream housing into premium market segments.
- Properties with 5 or more bathrooms command a median valuation of approximately **₹5 Cr**, highlighting their concentration within Gurgaon's luxury housing market.

Luxury Market Profiling

What Differentiates the Most Expensive Properties?

The residential housing market is not homogeneous. A small proportion of properties account for a disproportionately large share of market value and represent the luxury segment of the market.

By comparing the most expensive properties against the broader market, we can identify the characteristics that distinguish luxury homes and better understand the factors

associated with premium residential valuations in Gurgaon.

In [108…
```
price_cutoff = df_sales['PRICE'].quantile(0.90)

df_sales['MARKET_SEGMENT'] = np.where(
    df_sales['PRICE'] >= price_cutoff,
    'Top 10%',
    'Remaining 90%'
)

print(f"Luxury Threshold: ₹{price_cutoff:,.0f}")
```

```
Luxury Threshold: ₹62,500,000
```

In [110…
```
luxury_profile = (
    df_sales
    .groupby('MARKET_SEGMENT')
    .agg({
        'PRICE':'median',
        'AREA':'median',
        'BHK':'median',
        'BATHROOM_NUM':'median',
        'PRICE_SQFT':'median'
    })
    .round(0)
)

luxury_profile
```

Out[110…

MARKET_SEGMENT	PRICE	AREA	BHK	BATHROOM_NUM	PRICE_SQFT
Remaining 90%	25000000.0	1975.0	3.0	3.0	11761.0
Top 10%	87000000.0	3979.0	4.0	5.0	23967.0

In [111…
```
premium_df = pd.DataFrame({
    'Metric': ['Price','Area','BHK','Bathrooms','PPSF'],
    'Multiplier': [
        85500000/22000000,
        4000/1995,
        4/3,
        5/3,
        27631/11501
    ]
})

premium_df = premium_df.sort_values(
    'Multiplier',
    ascending=True
)
```

In [113…
```
# Calculate Luxury Premium Multipliers

premium_df = pd.DataFrame({
    'Metric': ['Price', 'PPSF', 'Area', 'Bathrooms', 'BHK'],
    'Multiplier': [
        luxury_profile.loc['Top 10%', 'PRICE'] /
```

```
        luxury_profile.loc['Remaining 90%', 'PRICE'],

        luxury_profile.loc['Top 10%', 'PRICE_SQFT'] /
        luxury_profile.loc['Remaining 90%', 'PRICE_SQFT'],

        luxury_profile.loc['Top 10%', 'AREA'] /
        luxury_profile.loc['Remaining 90%', 'AREA'],

        luxury_profile.loc['Top 10%', 'BATHROOM_NUM'] /
        luxury_profile.loc['Remaining 90%', 'BATHROOM_NUM'],

        luxury_profile.loc['Top 10%', 'BHK'] /
        luxury_profile.loc['Remaining 90%', 'BHK']
    ]
})

premium_df = premium_df.sort_values('Multiplier')

plt.figure(figsize=(10,5))

bars = plt.barh(
    premium_df['Metric'],
    premium_df['Multiplier'],
    color='#C8D5B9',
    edgecolor='black',
    linewidth=0.8
)

plt.title(
    'What Makes Luxury Properties Different?',
    fontsize=14,
    pad=15
)

plt.xlabel('Luxury Premium Multiplier (Top 10% vs Remaining 90%)')

plt.grid(
    axis='x',
    linestyle='--',
    alpha=0.35
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

for bar in bars:
    width = bar.get_width()

    plt.text(
        width + 0.03,
        bar.get_y() + bar.get_height()/2,
        f'{width:.1f}x',
        va='center',
        fontsize=10
    )

plt.tight_layout()
plt.show()
```



Key Insights

- Luxury properties command a **3.5x higher median price** than the remaining 90% of the market, making price the most visible differentiator between luxury and non-luxury housing segments.
- Luxury homes are not only more expensive but also significantly larger and more exclusive. The top 10% of properties offer approximately **2x larger living areas** and command **2x higher price-per-square-foot values**, indicating that buyers are paying both for additional space and for premium location, quality, and prestige.
- Property configuration also plays a role in luxury positioning, with luxury homes offering **1.7x more bathrooms** and **1.3x more bedrooms** than the broader market. This suggests that luxury housing is characterized more by enhanced livability, spacious layouts, and premium amenities rather than simply increasing the number of bedrooms.

Locality Efficiency Map

Evaluating Value Across Localities

High property prices do not necessarily indicate poor value, just as low prices do not automatically imply affordability. A more meaningful assessment considers both property valuations and the amount of living space buyers receive for their investment.

This analysis compares localities based on their median property price and median property size to identify areas that offer stronger value propositions versus those that command premium pricing with relatively limited space.

In [114…

```
locality_efficiency = (
    df_sales.groupby('LOCALITY')
    .agg(
        listings=('LOCALITY','count'),
```

```
        median_price=('PRICE','median'),
        median_area=('AREA','median')
    )
    .reset_index()
)

locality_efficiency = locality_efficiency[
    locality_efficiency['listings'] >= 20
]
```

In [115…

```
median_price = locality_efficiency['median_price'].median()
median_area = locality_efficiency['median_area'].median()

print(f"Median Price: ₹{median_price:,.0f}")
print(f"Median Area: {median_area:,.0f} sq.ft")
```

```
Median Price: ₹25,000,000
Median Area: 2,160 sq.ft
```

In [118…

```
plt.figure(figsize=(10,6))

plt.scatter(
    locality_efficiency['median_price'],
    locality_efficiency['median_area'],
    s=80,
    alpha=0.75,
    color='#C8D5B9',
    edgecolors='black'
)

# Reference Lines

plt.axvline(
    median_price,
    linestyle='--',
    linewidth=1.5,
    alpha=0.8
)

plt.axhline(
    median_area,
    linestyle='--',
    linewidth=1.5,
    alpha=0.8
)

# Label Most Expensive Localities

top_localities = locality_efficiency.nlargest(
    4,
    'median_price'
)

for _, row in top_localities.iterrows():
    plt.annotate(
        row['LOCALITY'],
        (row['median_price'], row['median_area']),
        xytext=(5,5),
        textcoords='offset points',
        fontsize=8
```

```
    )

plt.title(
    'Locality Efficiency Map',
    fontsize=14,
    pad=15
)

import matplotlib.ticker as ticker

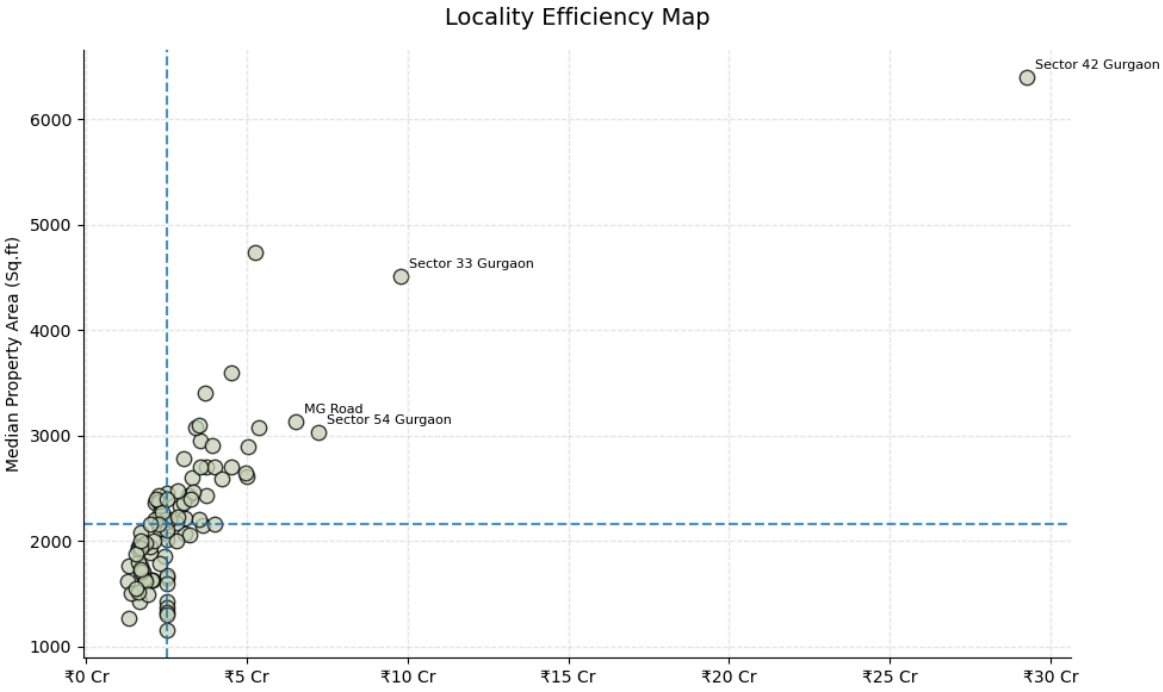
ax = plt.gca()

ax.xaxis.set_major_formatter(
    ticker.FuncFormatter(
        lambda x, pos: f'₹{x/1e7:.0f} Cr'
    )
)
plt.ylabel('Median Property Area (Sq.ft)')

plt.grid(
    linestyle='--',
    alpha=0.3
)

ax = plt.gca()
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.tight_layout()
plt.show()
```



Key Insights

- Most Gurgaon localities are clustered below the city medians for both property price and property size, indicating that the majority of the market operates within a relatively affordable and space-constrained segment. Only a small number of localities break away from this pattern and offer substantially larger homes.

- Sector 42 Gurgaon occupies a unique position in the market, combining exceptionally large median property sizes (≈6,400 sq.ft) with the highest median prices. This confirms its status as Gurgaon's premier ultra-luxury micro-market, where buyers pay a significant premium for both space and location exclusivity.
- Localities such as Sector 33 Gurgaon, MG Road, Sector 54 Gurgaon, and Sector 26 Gurgaon stand out for offering above-average property sizes without reaching the extreme price levels of Sector 42 Gurgaon. These markets may represent attractive opportunities for buyers seeking larger homes and stronger space-to-price efficiency than the city's most premium localities.

Strategic Recommendations

Based on the analysis of Gurgaon's residential housing market, the following recommendations can help buyers make more informed property decisions:

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1. Prioritize Property Area as the Primary Value Driver

Among all measurable property characteristics, area demonstrated the strongest relationship with property prices. Larger homes consistently commanded higher valuations across Gurgaon's residential market.

Recommendation: Buyers should treat property size as a primary evaluation criterion and compare homes on both total price and usable living space rather than focusing solely on budget or bedroom count.

---

2. Do Not Assume Expensive Localities Always Offer Better Value

Several premium localities command significantly higher price-per-square-foot values without delivering proportionally larger homes. In many cases, buyers are paying a premium for prestige, location advantages, exclusivity, and lifestyle benefits.

Recommendation: Before paying a locality premium, assess whether the additional cost is justified by factors such as connectivity, neighborhood quality, infrastructure access, and long-term appreciation potential.

---

3. Focus on Space-to-Price Efficiency

The locality efficiency analysis revealed that some localities offer substantially larger homes while remaining below the pricing levels of Gurgaon's most expensive micro-markets.

Recommendation: Buyers seeking long-term value should compare localities based on both property size and price. Markets offering above-average space at moderate prices may provide stronger value and future appreciation potential.

---

4. Evaluate Overall Livability, Not Just Bedroom Count

While BHK configuration and bathroom availability contribute to property valuations, the analysis suggests that buyers place greater importance on overall livability and functional space utilization.

Recommendation: Consider layout quality, room proportions, bathroom availability, and practical usability when comparing properties rather than relying solely on the number of bedrooms.

---

5. Define Your Purchase Objective Before Selecting a Locality

Different Gurgaon localities serve different buyer profiles. Some are optimized for affordability and family living, while others are positioned as premium or ultra-luxury markets.

Recommendation: Clearly identify whether your primary objective is affordability, larger living space, lifestyle quality, prestige, or long-term investment growth before shortlisting localities.

---

6. Use Locality-Level Analysis Instead of City-Wide Averages

Significant differences exist across Gurgaon localities in terms of pricing, property size, inventory concentration, and price-per-square-foot performance. City-wide averages often mask these variations.

Recommendation: Compare properties at the locality level rather than relying on overall market averages. Localized analysis provides a more accurate understanding of value and investment potential.

---

7. Recognize That Location Creates Pricing Power

The analysis showed that property size alone does not fully explain price differences. Several premium localities achieve substantially higher valuations despite offering homes of comparable size, demonstrating the strong influence of location and market positioning.

Recommendation: When evaluating a property, consider locality reputation, accessibility, surrounding infrastructure, and future development plans alongside

property specifications, as these factors can significantly influence long-term value creation.

Executive Summary

This analysis of **10,010 residential properties across 222 Gurgaon localities** uncovered important insights into the city's housing market, pricing dynamics, and locality-level value opportunities.

1. Gurgaon Remains Primarily a Mid-Market Residential Market

The market is largely centered around upper-middle-income housing, with a median property price of approximately **₹2.5 Cr**, a median property size of around **2,080 sq.ft**, and a median price-per-square-foot of roughly **₹12,300**. Most listings are concentrated within practical family-oriented housing segments rather than the ultra-luxury market.

2. Location Creates Significant Pricing Power

Substantial variation exists across Gurgaon localities, with premium micro-markets commanding price-per-square-foot values several times higher than the city median. Buyers frequently pay significant premiums for location advantages such as connectivity, prestige, infrastructure quality, and lifestyle amenities.

3. Property Size Is the Strongest Structural Driver of Valuation

Among all measurable property attributes, area exhibited the strongest relationship with property price. Larger homes consistently command higher valuations, although location and market positioning continue to play an important role in explaining price differences between similarly sized properties.

4. Luxury Housing Operates as a Distinct Market Segment

Luxury properties differ from the broader market not only in total price but also in property size, price-per-square-foot, and overall configuration. Premium localities such as Sector 42 Gurgaon demonstrate that exclusivity and location prestige can generate substantial pricing power beyond the impact of property size alone.

5. Meaningful Space-to-Price Opportunities Exist Across Gurgaon

The locality efficiency analysis identified several markets that offer substantially larger homes without requiring the extreme premiums observed in Gurgaon’s most expensive micro-markets. These localities may provide attractive opportunities for buyers seeking better value and long-term appreciation potential.

6. Locality-Level Analysis Is Essential for Informed Decision-Making

Significant differences exist between Gurgaon localities in terms of pricing, inventory concentration, property size, and valuation efficiency. Market-wide averages often conceal these differences, making locality-level evaluation critical for both homebuyers and investors.

Final Conclusion

Gurgaon's residential market cannot be understood through property price alone. Successful property decisions require balancing **location quality, living space, property configuration, valuation efficiency, and long-term growth potential**. Buyers who evaluate these factors together are better positioned to identify both suitable homes and attractive investment opportunities within Gurgaon's highly diverse housing market.

Download Tables for Dashboard Analysis

In [119…

```
df_sales.to_csv(
    r"D:\DATA analysis\projects\Key Projects\99 acre\Gurgaon Data\gurgaon_sales.
    index=False
)

print("Dataset exported successfully.")
```

```
Dataset exported successfully.
```

In [120…

```
df_rent.to_csv(
    r"D:\DATA analysis\projects\Key Projects\99 acre\Gurgaon Data\gurgaon_rent.c
    index=False
)

print("Dataset exported successfully.")
```

```
Dataset exported successfully.
```

In [ ]: